Tools and screeners

Trunk section: §6.2 Tools and screeners.

Talks to: Sources (§6.1: the TradingView scanner, and Alpaca and news for post-scan tests); the Card (§6.3), which receives every candidate; Registers (§6.6), which freeze each tool's photo and record the screener versions; Learning (§6.7), which judges every screener; Platform (§6.10), whose scheduler runs discovery.

Built on: METHODOLOGY (every tool, every screener and why), SCREENER-BUILDER (the definition, editor, validation, mirrors, how a scan is sent), TOOL-LIFECYCLE (adding, starting, pausing, stopping, deleting, the archive), IDEAS part 2 (the intent for the builder), MONITOR (the discovery schedule), PAGES (the suite and Manage tools pages).

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1. What a tool and a screener are

- **A tool** is one trading method, run as an experiment: its own screeners, its own photo minute and entries, its own history and model. Tools never share results, so one tool's outcome can never contaminate another's.
- **A screener** is one question put to TradingView's scanner: a set of rules, an order, a size, the minutes it may add cards, and usually a **mirror** (the same setup facing the other way).
- **Discovery** runs each due screener and turns its matches into candidates for the card engine.

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2. Rules every tool keeps

2.1 The tradability floor

Added to **every screener on every tool** at the moment a scan is sent, not written into any screener, so one setting changes all:

Rule	Why
price ≥ $1	below it the spread eats the trade
average volume ≥ 1M shares a day	below it you cannot get out at a price you chose
ATR ≥ $1	not enough movement to pay for the risk
ATR ≥ 3 % of price	the same question relative to the stock (TradingView cannot compare two columns, so this leg runs locally after the scan)

Settings `minPrice`, `minAvgVolume`, `minAtr`, `minAtrPct`; 0 switches a leg off. **Two named exceptions:** the neglected-name screeners (T6 Unexplained Move and its mirror) keep only the price floor; a **list-only** screener takes no floor and its matches never become cards.

2.2 How success is measured

No stop, no target, no exit: the question is whether a screener finds stocks that move. From each entry minute to the close, `upR = (hh − entry) ÷ ATR14` and `downR = (entry − ll) ÷ ATR14`; a good move is `upR ≥ 1.3`; ATR14 from daily bars strictly before today; regular session only. The opening range (09:30–09:35), the minutes from the open at discovery, and whether another tool shortlisted the card are recorded for every card (`REGISTERS.md` (to write)).

2.3 The photo

Each tool takes **one photo a day** (R1) of every card on screen, one minute after one of its discovery scans, so it freezes a fresh list. A card found after the photo can be traded but teaches nothing.

2.4 Discovery and refresh

A screener’s run window gates **discovery only**. Cards already on screen are re-quoted every 5 minutes, 04:00–16:00, whatever the windows.

2.5 Mirrors are made, not flipped

Rule kind	Mirrors to
directional field (change, change from open, gap, pre-market change and gap, performance W/1M/3M/6M/Y) above a number	operator flips **and** the number changes sign: `change > 5` → `change < −5`
a range on a directional field	negated: `3…10` → `−10…−3`
RSI (0–100)	reflected around 50: `RSI > 70` → `RSI < 30`
two columns compared	operator flips and each side swaps to its counterpart: `close > 1-month high` → `close < 1-month low` (also 3-month, 52-week, day high/low)
quality guards (price, volume, cap, float, …), equals, has, non-directional between	unchanged

The sort flips when it is on a directional field. A mirror keeps the same windows, because the pair must be measured over the same minutes.

2.6 The benchmark

Stocks in Play (T9): `close > 5`, the floor, sorted by relative volume, top 20. No pattern, no direction, no mirror, and nothing structural may ever be added to it (a test enforces this). Every screener must beat it by a clear margin (5 points). In the one strong published day-trading result, the edge came almost entirely from the relative-volume ranking (Sharpe 2.81 with it, 0.48 without), so it is a hard benchmark.

2.7 Order of work

1. **Collect** for about a month. 2. **Cut** the screeners that do not find movers. 3. **Then setups**: entries, stops and targets, only for the survivors.

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3. The tools, one by one

Eleven tools registered; five running today. Windows are New York time.

Tool	Name	Status	Photo · entry A · entry B
T1	Swing trades after huge volume (the control)	running	09:36 · 09:37 · 09:40
T2	Momentum	running	09:46 · 09:47 · 09:51
T3	Gappers	archived	09:36 · 09:37 · 09:40
T4	VWAP Reclaim	archived	10:01 · 10:02 · 10:06
T5	20-Day Break	archived	09:46 · 09:47 · 09:51
T6	Unexplained Move	running	10:16 · 10:17 · 10:21 (D-S9)
T7	Liquid Movers	running	09:36 · 09:37 · 09:40
T8	CANSLIM (split into T10 and T11 on 2026-09-04)	archived	09:46 · 09:47 · 09:51
T9	Stocks in Play (the benchmark)	archived (D-S14)	09:36 · 09:37 · 09:40
T10	Growth Stock Breakout (stopped 2026-09-21)	archived	09:46 · 09:47 · 09:51
T11	Growth Stock Pullback	running	09:46 · 09:47 · 09:51

T1 · Swing trades after huge volume: the control

The only tool with real history, and the one measured edge so far. All three screeners run all day, on purpose.

```
Trend       close ≥ 20 · close ≥ SMA5 · close ≥ VWAP · close|1W > VWAP|1W ·
            close|1M > VWAP|1M · EMA50|1 > EMA120|1 · close|1 ≥ EMA50|1 ·
            VWAP ≥ SMA75|5 · relative_volume_intraday|5 > 3 ·
            relative_volume_10d_calc > 1.5 · average_volume_90d_calc > 1M
Pre-Mkt     premarket_volume > 1.5M · relative_volume_10d_calc > 3 ·
            average_volume_10d_calc > 2M · close ≥ 1
Big Move    relative_volume_10d_calc > 10 · close ≥ 2 · average_volume_10d_calc > 2M
```

The measured edge: the overlap of Big Move and Pre-Mkt made a good move on 54 % of cards against 13 % for everything else (Fisher p = 0.0015, over 11 days, surviving the removal of the two biggest winners). It is read off the 09:36 photo; giving Pre-Mkt the window its filters suggest would strip the tag from every frozen row and delete the measurement. **T1 stays exactly as it is**, so the other tools have something to be compared with.

T2 · Momentum

A clean daily uptrend breaking to a new monthly high.

```
Breakout    SMA5|1 > EMA9|1 > EMA13|1 > EMA20|1 · close ≥ High.1M · close ≥ 1 ·
            average_volume_10d_calc > 500K
Mirror      the stack inverted · close ≤ Low.1M
Window      09:00–10:00
```

The stack says the trend; the break makes it an event. Positions are closed by 10:30, so anything found after 10:00 is a candidate nobody can take. The photo at 09:46 lets a one-spike opening break fail first.

T3 · Gappers (archived)

```
Gap up      premarket_change > 5 · premarket_volume > 500K ·
            relative_volume_10d_calc > 2 · close ≥ 1 · float_shares_outstanding < 50M
Mirror      premarket_change < −5
Window      04:00–10:30
```

A gap with no pre-market volume is a quote; the small float lets a gap extend. Pre-market columns freeze at the bell, hence the window.

T4 · VWAP Reclaim (archived): the reversal tool

```
Reclaim     close crosses_above VWAP · relative_volume_10d_calc > 2 · close ≥ 1 ·
            average_volume_10d_calc > 1M
Mirror      crosses_below
Window      09:45–15:30
```

Reversals come late. **Known limit:** a cross is a moment, and discovery runs every 15 minutes after 10:00, so some crosses are missed between scans.

T5 · 20-Day Break (archived)

```
Break       close ≥ High.1M · relative_volume_10d_calc > 1.5 · close ≥ 1 ·
            average_volume_10d_calc > 500K
Mirror      close ≤ Low.1M
Window      09:30–16:00
Kept, off   52-Week Break: close ≥ price_52_week_high · market_cap_basic > $1B ·
            relative_volume_10d_calc > 1.5 · average_volume_10d_calc > 1M
```

It replaced the 52-week break, which produced one frozen day in the whole archive. “With supporting news” is a card filter, not a screener column.

T6 · Unexplained Move: the only mean-reversion tool

```
Net          change > 10 · close ≥ 1        (a wide net, not the rule)
Mirror       change < −10
Window       11:30–16:00
Kept, off    Overextended: RSI > 70 · close ≥ EMA20 · relative_volume_10d_calc > 3 ·
             close ≥ 1 · average_volume_10d_calc > 1M (mirror RSI < 30, close ≤ EMA20),
             window 10:00–16:00
```

The real rules run after the scan, because the scanner cannot express them:

Test	Data	Rule
Move rate	Alpaca 1-minute bars today	≥ +15 % (mirror ≤ −15 %) over the last **120 minutes**, close to close; no full window = dropped. (TradingView's `change|120` is a fixed candle and splits moves across its boundary)
No news	the full news stack	any catalyst = dropped; a failed lookup = dropped
Six-month trend	Alpaca daily closes **before today**, 126 sessions	base: dropped if the six months rose more than 20 %; mirror: dropped if they fell more than 20 %; under 126 sessions = dropped

Survivors carry `move2hPct` and `trend6mPct`. Every drop is counted with its reason; if the gate itself throws, candidates pass untouched (an empty scan would look like a quiet market). Floor: price only (neglected names). **Blind spot:** a stock down 10 % in the morning and up 15 % since lunch is flat on the day and never reaches the net.

T7 · Liquid Movers

Two screeners for two sessions, **not** a mirror pair. Both take the top 25.

```
Pre-Market Gap    gap not between −3 % and +3 % · ATR ≥ $1 ·
   04:00–09:30    average_volume_90d_calc ≥ 2M · premarket_volume > 1M · close > 5
After Open Volume average_volume_10d_calc > 5M · relative_volume_intraday|5 > 3 ·
   09:30–16:00    volume > 10M · change not between −3 % and +3 % · close > 5
```

History: it once returned 75 names a day because both volume rules grew through the session, the floor was $1, the gap screener did not require any pre-market trade, and each took the top 50. All four were fixed. `relative_volume_10d_calc` was removed (day-to-date volume over a full-day average means a different thing each hour). The pre-market screener feeds training; the after-open one finds names after the photo.

T8 · CANSLIM (archived) → T10 and T11

```
Breakout    earnings_per_share_diluted_yoy_growth_fq ≥ 25 ·
(T8, T10)   earnings_per_share_diluted_yoy_growth_fy ≥ 25 ·
            total_revenue_yoy_growth_fq ≥ 15 · close ≥ price_52_week_high ·
            Perf.6M > 30 · relative_volume_10d_calc > 1.5 ·
            total_shares_outstanding_fundamental < 1B · market_cap_basic > $300M
Pullback    the same company quality · close ≥ SMA50 · close ≤ EMA20
(T8, T11)   (T11: without the revenue, relative-volume and share-count rules;
            window 09:30–16:00, top 50)
```

Split because T8 ran all three screeners in one database, so no register could say which one produced a result. **What a screener cannot be, honestly:** I (no dependable column: left out), M (describes the market, not a stock: not a filter, on purpose), L (Perf.6M is a threshold, not a percentile: strength, not leadership). The full O'Neil reading is on every card instead (ONEIL.md). T8's shared ★ CANSLIM list (90-day membership) was retired on 2026-10-02.

T9 · Stocks in Play: the benchmark (archived)

`close > 5`, the floor, sorted by relative volume, top 20 (§2.6).

The unified shortlist

Each tool keeps its own shortlist; anything shortlisted on any tool shows on every other tool's card as `★ also T1 T7`, never touching that tool's own star or model (`REGISTERS.md` (to write)).

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4. The legacy builder

4.1 A screener is a row

Field	Meaning
`key`	stable id from the name at creation; stamped on every matched card; never changes, even on rename
`name`	shown; renamed freely
`filters`	rules `{ left, operation, right }`, TradingView's own shape
`sort`, `limit`	default change % high to low, 50
`runFrom`, `runTo`	the run window; empty = every scan
`checkFrom`, `checkTo`	when it is worth opening the tool; empty = the run window
`mirrorOf`	the screener it mirrors
`labelOnly`	a list screener: matches never become cards
`enabled`, `pausedAt`, `pausedReason`	on/off and why

`right` is a number, a pair, or **another column** (that makes "close above EMA 20" expressible). A column may carry a timeframe after `|`: `EMA50|1` daily, `close|1W` weekly, `relative_volume_intraday|5` five-minute.

15 operators: above, above or equal, below, below or equal, equals, not equal, between, not between, crosses, crosses above, crosses below, above by %, below by %, has, has none of.

52 columns offered:

Group	Columns
Price	close, open, day high, day low, change %, change from open %, gap %, VWAP
Averages	SMA 5, 20, 50, 200; EMA 9, 13, 20, 50, 120, 200
Ranges	1-month high/low, 3-month high/low, all-time high, 52-week high/low
Volume	relative volume (10-day), relative volume (5-minute), volume, average volume 10-day and 90-day, pre-market volume, pre-market change %, pre-market gap %
Size and short	market cap, float shares, short % of float, shares outstanding
Movement	ATR, performance week / month / 3-month / 6-month / year, RSI
Fundamentals	EPS growth (quarter, year, TTM), revenue growth (quarter, TTM), ROE
Labels	sector, exchange

TradingView **silently ignores a column it does not know**, which widens the screener instead of failing; `scripts/verify-tv-fields.js` checks every offered column.

4.2 The editor and the list

The list shows each screener's rules in words, its window on a timeline with the tool's capture minute and why it was chosen, the floor as one shared line, and badges (Paused, List only, Mirror of …). Each has Rename, Mirror, Pause / Resume, Delete. The editor has name, rules (field · condition · number or column), the floor shown read-only, sort, max results, run window, and **Test on TradingView**. The check window, list-only and the mirror link are not in the editor.

4.3 Validation and test

Refused on save, with every reason: no name or rules; unknown column, operator or missing value; between without exactly two values; unknown sort column; a window with one end, a time not `HH:MM`, or a start not before its end. Strict, because a malformed rule returns an empty list that looks like "nothing matched today".

Test runs the definition as it would really run, floor included, without saving: `count` (capped at the limit), `totalCount` (how many matched in all: the real measure of selectivity), time taken, a sample of 10. **Preview mirror** shows a mirror without saving.

4.4 Rename, pause, delete, seeding

Rename keeps the key (and re-points any mirror). Pause stops new matches and touches nothing found; the pause time is set when paused. Delete removes the row; cards keep a key nothing explains (pausing is safer). A new tool gets its seed set from `seedScreeners.js`, or nothing. **Startup repairs** run keyed, one-time fixes on every start (D-S10). `screenerSummary.js` turns the live rules into words, so a description cannot drift from what runs.

5. Legacy discovery

Schedule (New York, weekdays): every 30 minutes 04:00–09:00, every 5 09:00–10:00, every 15 10:00–16:00, once at 16:00. A scan is skipped when the tool is paused or another run is going.

Each due screener (enabled, window containing the minute) is one request, in parallel, to `POST scanner.tradingview.com/america/scan`:

Part	Value
`filter`	the screener's rules **plus the floor** (added at the server, so the top N slots are not spent on ineligible stocks)
`filter2`	the base universe: common stock, preferred stock, depositary receipts, funds that are not ETFs, mutual or closed-end; never pre-IPO
`columns`	the 32 card columns (CARD.md §2.4)
`sort`, `range`	the screener's sort; `[0, limit]`
headers	a browser User-Agent, Origin and Referer of tradingview.com; 15 s timeout

Checks on the answer: misaligned columns → that screener returns nothing this scan; week = month (or month = quarter) range on every one of at least 20 rows → logged once; the ATR % leg applied locally; a failed screener is logged and the rest continue. Then the T6 gate (§3), then **merge**: one row per ticker per tool, `screenerKeys` listing every screener that matched, a field empty in one match filled from another, never overwritten.

6. Legacy lifecycle

Each tool is its own program: pm2 process `tool-Tn`, its scorer `scorer-Tn` on port + 1, database `data/tn.db` (T1: `data/tradedesk.db`), model folder, temp folder, memory cap (T1 280M; T2, T6 240M; T11 200M; T7 180M; others 140M; scorer 180M; archive 120M). T1 also serves the landing page and Manage tools, so it can be neither stopped nor deleted.

The registry merges `tools.config.json` (built-in tools) and `data/tools.local.json` (tools added from the page, and overrides of name, description, colour, capture times, enabled, deleted, pendingRestart). Flags: `enabled`, `archive` (served read-only when stopped), `scorer`.

Adding a tool: at most 6 added tools; id one past the highest ever used (never reused); port the next free multiple of ten; no scorer; capture 09:36 / 09:37 / 09:40 by default; started at

once if at least **180 MB** is free. Capture times must run `r1 < entryA < entryB`, `r1 ≥ 09:30`, `entryB ≤ 15:50`.

States and buttons: Running ⇄ Paused (the program stays up, stops scanning; stored in the tool’s database, survives a restart); Stop (process deleted, the **archive** serves its registers on its port); Start; Delete (hidden, history still served); Restore (back as stopped). **The session rule:** a capture-time change on a running tool between 04:00 and 16:30 is applied at 16:50, because a restart empties the list built since 04:00.

The archive: one read-only process serving each stopped tool’s `/api/warehouse/*` on that tool’s own port, so qp sees no difference.

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7. What failed

ID	Fault	Root cause	Prevented by
D-S9	T6 photographs at 10:16 a screener whose window opens at 11:30: its photo holds nothing T6's real screener found, so it trains on nothing	capture time inherited from the screener it replaced; times and windows never checked together	checked on save (§8.3)
D-S10	startup repairs guarded by "the field is empty" put back a window the trader cleared or a rule they removed, silently	one-time fixes run as start-up code	versioned migrations that run once, ever
D-S14	with T9 stopped, the comparison page has no benchmark	the benchmark is a whole process costing ~90 MB	tools as rows: the benchmark costs a few MB and always runs
—	editing a rule changes what a key means for every past card	no versions	screener versions; R1 records them
—	each tool is a whole program (~90 MB, plus a scorer); tools stopped for lack of memory	one process per tool	one discovery engine for all tools
—	the check window, list-only and mirror link cannot be set in the editor	editor incomplete	every field editable
—	post-scan tests are a hidden gate keyed on two screener keys	tests outside the definition	tests are named steps of the screener
—	a rule that can never match, or matches everything, is saved without warning	no selectivity check	Test runs on save and shows `totalCount`
—	T4 misses crosses between 15-minute scans	a cross is a moment	a "crossed since the last scan" test from bars, when wanted
—	T6 cannot see a morning drop followed by an afternoon rise	the net is the day's change	a net on the 2-hour move itself (from bars), when wanted

8. The new design

8.1 Tools are rows

Column	Meaning
`id`	T1, T2, …; never reused
`name`, `description`, `accent`	shown
`status`	`running`, `paused` (with when and why), `archived`, `deleted`
`capture_r1`, `entry_a`, `entry_b`	New York `HH:MM`, with a `why` note
`role`	`method`, `control` (T1), or `benchmark` (T9)
`created_at`	—

One discovery engine runs every **running** tool. An archived tool keeps its history in the same database and is readable everywhere; there is no archive process and no port per tool. A paused tool keeps its cards on screen and finds nothing new. **The benchmark always runs**, because it now costs a few MB.

8.2 Screeners are versioned

Table	Holds
`screeners`	`tool`, `key` (stable), `name`, `status`, `mirror_of`, `purpose` (`cards` or `list`)
`screener_versions`	`screener`, `version`, `rules`, `sort`, `limit`, `run_window`, `check_window`, `floor` (`standard`, `price_only`, `none`), `post_tests`, `created_at`, `note`

Saving a change creates a new version; the old one is kept. Each tool card records the version of every screener that matched, and R1 freezes it, so two definitions are never mixed under one name. Pausing and resuming do not create versions; renaming does not either (the key is stable).

Post-scan tests are part of the version, each with its parameters:

Test	Parameters	Data
`move_rate`	minutes (120), threshold (+15 %, or −15 % for a mirror)	1-minute bars
`no_news`	days	the card’s news facts
`trend_guard`	sessions (126), limit (20 %), side	daily closes before today

The builder shows them as rules under the scanner rules, labelled “applied after the scan”, so what the scanner cannot express is said, not faked.

8.3 Checked on save

Refused, with every reason:

- everything §4.3 checks;
- an unknown column, checked against the adapter's live column list (not a hand-kept list);
- **the capture minute is before every running screener's run window has opened** (the D-S9 case);
- `r1 < entryA < entryB`, `r1 ≥ 09:30`, `entryB ≤ 15:50`;
- a mirror whose windows differ from its pair.

Warned (saved, with the warning shown): Test returns `totalCount` 0, or more than 2,000.

8.4 Discovery

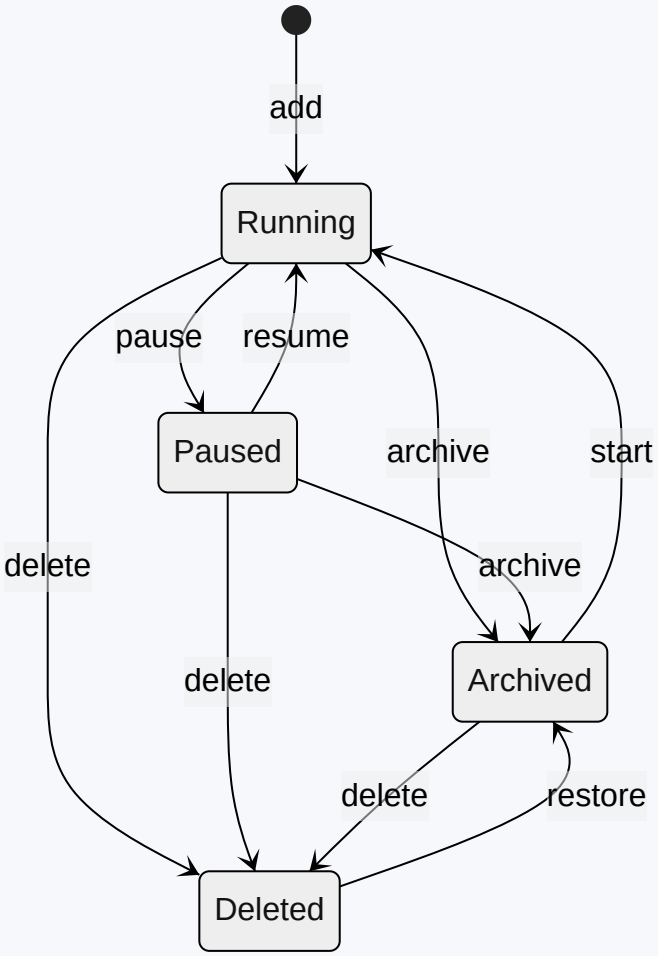
Run by the platform scheduler on the legacy cadence. For each due screener version of each running tool:

1. one request through the TradingView adapter (§5), floor added at the server, 32 columns;
2. the adapter's checks (alignment, identical ranges);
3. the ATR % leg applied locally (unless the floor says otherwise);
4. post-scan tests, each drop counted with its reason;
5. merge into candidates: tool, ticker, screener key and version, match time;
6. hand new tickers to the card engine; update tool cards.

One request per screener per scan, whatever the number of tools. Two tools with an identical screener version share the request.

A **scan report** is stored per tool per scan: screeners asked, rows returned, drops by reason, candidates, new cards, time taken, errors.

8.5 Lifecycle



Every state change is a row change, logged with who and when. A capture-time change takes effect at the next trading day (or at once outside 04:00–16:30), because the photo of a day must use one time; no restart is involved.

8.6 The pages

Page	Shows
Suite	every tool: status, today’s counts, photo time, last scan
A tool’s Screeners tab	each screener with its rules in words, versions, window timeline with the capture minute, floor, post-scan tests, badges; the full editor (every field, including check window, purpose, mirror link)
Manage tools	add, edit (with the checks of §8.3), pause, resume, archive, start, delete, restore
Comparison	every screener against the benchmark and its mirror (`LEARNING.md` (to write))

8.7 Seeds and migrations

Seed screener sets become **versioned migrations**: each runs once, ever, and is recorded. A migration never overwrites a field the trader changed after the seed.

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9. Contracts

Owns

Table	Key
`tools`	id
`screeners`	tool, key
`screener_versions`	screener, version
`scan_reports`	tool, scan instant

Writes to others

To	What
card	candidates: tool, ticker, screener key and version, match time, test results (`move2hPct` , `trend6mPct`)

Read by

Domain	What
registers	capture and entry times; screener versions for R1
learning	screeners, mirrors, benchmark role
platform	the discovery schedule
quant	tools and their history (backtest universes)

Routes (desk-web)

Route	Does
`GET /api/tools`, `POST /api/tools`, `PATCH /api/tools/:id`	list, add, edit (checked)
`POST /api/tools/:id/{pause,resume,archive,start,delete,restore}`	lifecycle
`GET /api/tools/:id/screeners`, `POST …`, `PATCH …/:key`	list, add, edit (a new version)
`POST /api/screeners/test`, `POST /api/screeners/preview-mirror`	test without saving
`GET /api/screeners/meta`	columns (from the adapter), operators, post-scan tests
`GET /api/tools/:id/scans?day=`	scan reports

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10. How the rebuild is proven

1. **Request parity.** For one day, the new discovery engine sends exactly the same request bodies as each old tool (recorded and compared).
2. **Candidate parity.** For a week, each tool's candidates and tool cards match the old tool's R0, screener keys and first-seen times included.
3. **Mirror tests.** Every row of §2.5 has a fixture, including the high/low counterpart swap.
4. **The D-S9 check.** Saving T6's current capture time with its 11:30 window is refused, with the reason.
5. **Benchmark guard.** Adding a structural rule to the benchmark tool is refused (the test carried over).